FORMOSA CRYPTO

Completing the Chain:

Verified Implementations of Hash-Based Signatures and Their Security

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Repository

Paper

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Jasmin [ABB+17, ABB+20]

- Stack zeroization [OBG+24]
- (Static) Safety checker based on abstract interpretation
- Compiler correctness proved in Rocq
- Extraction to EasyCrypt

Compiler Correctness & Constant-Time Preservation

Compilation preserves the semantics of safe source programs and constant-time behavior: functional correctness and constant-time guarantees established at source level carries to the assembly level.

QUESTIONS?

References I

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High-assurance zeroization.

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